

From architecture to a tested pilot

R&D;, funding, people and the next decisions

A practical plan for trusted civilian information during prolonged communications outages.

10-12 weeks	EUR 30,000	EUR 5,000
Proposed demonstrator	Planning cash budget	Initial allocation within total

Combines the complete technical assessment, the Latvian IDC response draft and the partner network discussed with Kristaps.

Includes VUGD, Valsts krīzes centrs, IDC, senior engineers, RTU students, Make Riga, Startschool, JCI Latvia, Ukrainian community organisations, the Austin MBA connection, Andrejs, Inga Ulmane and a prospective radio lead whose name may be Dzintars.

8 September 2026

Proposed roles and planning estimates. Contact details and access were supplied by Kristaps; commitments and institutional endorsements remain to be confirmed. No physical radio or phone tests were performed in this review.

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The planning section is intended for project coordination. Individual relationships and proposed asks should be confirmed before adapting material for external circulation.

Delivery, funding and network plan

1. The next decision and the first deliverable

Recommendation: organise a 10-12-week demonstrator with one operational partner, one accountable technical lead and a planning budget of EUR 30,000 excluding VAT. Allocate approximately EUR 5,000 to initial feasibility work, then commit the remainder against measured results. The first allocation is part of the EUR 30,000 total.

The demonstrator should show this complete journey: an authorised test publisher creates a signed update; a radio gateway receives it; a phone verifies it and updates its downloaded map; another phone receives it during a nearby exchange; a resident creates an encrypted report; that report reaches the centre later; a verifiable receipt returns to the originating phone.

The first operational scenarios should concern prolonged outages, resource availability and delayed assistance reports. Delivery through future encounters has no guaranteed arrival time. Any role in time-critical warnings needs separately established delivery requirements alongside official warning channels.

This document brings together the technical research, proposed next steps, funding research and all relationships named in this conversation. Appendix A retains the full technical assessment and its limitations. Appendix B contains the Latvian IDC response draft. Source links are collected at the end.

Evidence and commitment status. The technical findings describe the code revisions reviewed on 8 September 2026; they do not verify a later deployment. No physical Atbalsts radio, acoustic, range, battery or background-phone measurements were performed. Budgets and thresholds are proposals. Contacts and access were described by Kristaps; no new commitments, funding awards or institutional endorsements were secured in this conversation. No outreach has been sent. Personal names and roles should be confirmed before external use; Dzintars's name is explicitly uncertain.

The first month should produce a named integration lead, a radio lead, a candidate test site, an agreed institutional scenario, an initial funding commitment and a recorded feasibility result. The wider network supports those deliverables through bounded assignments, equipment, facilities, introductions and supervised testing.

2. Architecture to carry into the pilot

The baseline is a locally installed application with durable storage, preloaded reference data and a shared message format that remains verifiable across different communication paths. External LoRa radios connect local gateways. Phones reach gateways through tested local Wi-Fi or BLE interfaces. Nearby phone exchange starts with the app open and explicit user action.

Layer	Pilot responsibility	Main limitation to test
Publishing and regional distribution	Approve, version and sign immutable bulletins; provide a route into the region and predelegated local publishing where needed	An isolated local mesh cannot obtain a national update without an independent feed or physical carrier
LoRa and community gateway	External radio, persistent queues, local phone access and backup power planning	Range, airtime, interference, upstream capacity and energy use depend on the installation

Layer	Pilot responsibility	Main limitation to test
Phone and offline map	Verify messages, apply event revisions, show source/age and preserve private outgoing reports	Ordinary phones have no usable LoRa receiver; map and trust material must be installed before the outage
Nearby exchange	Exchange missing signed objects and encrypted reports; deduplicate and carry them onward	Foreground exchange is the baseline; background and locked-phone behaviour require device-specific evidence
Audio experiment	Decode audible modem data from a separate radio receiver into the same signed objects	A one-way broadcast path; decoder success, broadcast compatibility and airtime are unmeasured

Security belongs above every transport. Official messages are signed by authorised publishers; relays carry no authority signing keys. Private reports are encrypted for their recipient. Version, expiry, cancellation, deduplication, quotas and bounded queues contain stale-state and flooding failures. The application distinguishes saved locally, copied to a relay, received by the centre and reviewed/acted on.

A compromised relay can still delay, drop or flood traffic. Signatures cannot prevent jamming or establish the truth of a citizen observation. Test failure visibility and recovery as well as successful delivery. Appendix A contains the detailed threat model and proposed tests.

Keep the audio work independent of the core build. The 10-30 bit/s figure remains an unverified application-throughput assumption. Preapproved text and map references can be selected by compact codes; pictures and video wait for a faster link. Near-ultrasonic sound is not the initial broadcast choice.

3. Institutional relationships and first commitments

Use the existing questions from Iekšlietu digitālais centrs as the starting point for a jointly reviewed pilot brief. The first institutional meeting should settle three use cases, who may publish the test information, who receives incoming reports, which site is realistic and what evidence would justify further work.

Relationship	Proposed contribution	First bounded ask
Artūrs Pokšāns / Iekšlietu digitālais centrs	Requirements counterpart and technical evaluation	Review the proposed end-to-end scenario, nominate a participant and help identify an operational pilot owner
VUGD contacts	Practitioner requirements and exercise observation	One practitioner to review resident reports, information wording and the exercise; identify relevant existing workflows
Valsts krīzes centrs leadership	Institutional coordination and sponsorship of the evaluation process	Nominate a counterpart; clarify publication authority and the organisation that could own a subsequent pilot
One municipality, to be recruited	Test location and local operations	Two usable locations, a civil-protection contact, power/access arrangements and roughly 10-20 initial exercise participants

Relationship	Proposed contribution	First bounded ask
Mentor already consulted	Critical review and introductions	Review the two-page brief; introduce a decision-maker, a senior engineer and a possible sponsor

Aim to record the named person, contribution, availability, next date and limits of each commitment. A pilot participation letter should describe the site, staff time or evaluation role actually offered. Interest in the project is not evidence of a grant award or purchase commitment.

Kristaps should coordinate a 60-minute initial session and circulate the resulting brief. Include the message journey, operator responsibilities, test/real-message separation, a proposed exercise site and measurable success criteria. Ask who would handle resident reports outside office hours and who could fund continued operation after a pilot.

4. Technical advisers and delivery leadership

Relationship	Proposed role	First bounded ask
Senior hackers / experienced technical contacts	Integration leadership and independent technical challenge	One person to estimate and own the complete implementation; a different reviewer to challenge security and test results
Dzintars - name to confirm	Candidate radio engineering lead and radio-community mentor	A four-week feasibility assignment: equipment/configuration review, a few representative routes, one controlled field test and supervision of two contributors
Andrejs - described by Kristaps as founder of Tildes birojs	Senior product/technical adviser and introductions	A 45-minute architecture and pilot review; an introduction to an experienced implementation reviewer or engineer; assess a possible bounded contribution
RTU students and a supervisor to be recruited	Measurements, tooling and supervised research	A small team with named assignments, a reviewer and a demonstration deadline
Make Riga hackerspace	Practical radio/hardware sessions	Confirm bench access, inventory possible loan equipment, find experienced operators and organise two build/test sessions

Confirm native iPhone/Android integration experience and practical RF measurement experience separately. Enthusiasm and general coding skill do not by themselves cover those responsibilities. The integration lead owns software interfaces and release readiness; the radio lead owns radio configuration, antenna/power choices and measurements. They jointly demonstrate the complete chain.

Andrejs and a possible Tilde contribution. Explore approved multilingual instruction packs, consistent crisis terminology and comprehension testing. Compact event codes should select approved wording already installed on the phone. Version the dictionaries and define behaviour when a device lacks the required package. Ask what

advice, staff time or sponsorship is realistically available; do not assume an organisational commitment from a personal introduction.

Dzintars and the radio community. Ask him to identify what can be borrowed, recommend a small number of configurations, check permitted operating conditions with the relevant expertise, and document delivery probability, latency, capacity in both directions and energy use. Pair experienced operators with RTU or Startschool contributors who build logs and dashboards. The community can also develop future installation, maintenance and exercise skills.

For research access beyond existing contacts, [RTU's Industry Partnership Hub](#) ^[1] provides a route to suitable researchers and facilities. For a later authorised broadcast experiment, [LVRTC's radio services](#) ^[2] cover broadcast engineering and measurements. Neither access nor free airtime is assumed.

5. Community, international connections and recruitment

Relationship	Proposed contribution	First bounded ask
JCI Latvia	Event organisation, business introductions and volunteers	One organiser and three introductions to companies that could fund a milestone, lend equipment or contribute staff time
Startschool	Practical coding challenges and recruitment	Run a small supervised challenge; recruit contributors who produce reviewable work into the pilot team
Ukraine-support organisations and Ukrainians in Latvia	Experience of communication failures and usability	Four to six voluntary interviews, then a small group reviewing the journey using fictional incidents
Partner studying for an MBA in Austin for six months	Sponsorship, research introductions and purchasing evidence	During the first month: ten relevant conversations, two tailored sponsor proposals, one research introduction and a short buyer/procurement assessment
Inga Ulmane - described by Kristaps as a diplomat representing LIAA in Austria	Introductions to programme owners and Austrian partners	Introduce the relevant Latvian funding adviser and one Austrian applied-research or civil-protection organisation; assess later consortium opportunities

Ukrainian participants can surface failures involving depleted batteries, shared phones, confusing instructions, language, information trust and uncertainty over whether a request arrived. Use voluntary participation and fictional exercise data, compensate participation where possible, and record practical design findings without soliciting sensitive operational information. Lived experience informs the design; it is not proof that a particular radio architecture works.

Give the Austin partner a concise offer: maintaining trusted civilian information during prolonged communications outages, with a defined Latvian test setting and measurable results. Seek funding of engineering time, relevant research advice and evidence of purchasing requirements. Do not assume a particular university, laboratory or sponsor is already available through the MBA.

Give Inga the reviewed brief, actual partner contributions, the budget and two specific introduction requests. A subsequent Austrian evaluation could provide a second environment or a role in a European research consortium. LIAA programme owners assess eligibility; an introduction is not funding approval.

Start with Andrejs and the senior engineers reviewing the technical brief, then refine it with VUGD/IDC/crisis-centre contacts. Use that version for Inga's introductions and the Austin sponsorship conversations. Confirm full names and role descriptions before putting personal contacts into external partner materials.

6. How to organise the R&D

Kristaps owns scope, institutional relationships, funding and the decision log. One paid integration lead owns the backlog and complete demonstrator. The radio lead owns the RF work. A separate senior reviewer challenges the security model and results. One community organiser handles events and recruitment. These are proposed roles; assignments and availability must be agreed.

Work package	Accountable role	Reviewable output
Shared protocol, publishing and receipts	Integration lead with security review	Signed immutable event objects, version/cancellation rules, private report envelopes and recipient-verifiable receipts
Native offline app	Mobile engineer under integration lead	Cold start without cloud dependencies; downloaded map; durable queues; clear data age and delivery states
Radio and gateway	Radio lead plus gateway engineer	Pinned configuration, durable priority queues, tested local connection, field logs and power/restart results
Nearby exchange	Mobile engineer with device-test contributors	Foreground iPhone/Android exchange of missing objects and delayed reports, including incompatible-device results
Audio feasibility	Radio/DSP contributor with a senior reviewer	Time-bounded bench study, then an authorised broadcast-chain experiment only if justified by the first results
Independent validation	Reviewer who did not implement the relevant control	Threat-model findings, malformed/replayed message tests, false-receipt rejection and operational walkthrough

Run a weekly technical demonstration, a short weekly coordination meeting and a monthly institutional review when counterparts can participate. Each experiment records a hypothesis, equipment/configuration, software revision, conditions, sample size, observed failures and the resulting decision. Keep a separate list of unproven claims.

Agree code, data and publication rights before collaboration begins. Keep repositories, contribution review and test data organised around Atbalsts. Record borrowed equipment and donated time separately from cash spending. The lead reviews security-sensitive changes and controls releases; students contribute through bounded assignments with explicit acceptance criteria.

Useful student/challenge tasks include offline package validation, message-age UI, phone compatibility testing, a radio log collector, a gateway-status dashboard with synthetic data, audio-noise experiments and documentation

of repeatable exercises. A first task should fit roughly one to three working days and end in a reviewed change or usable measurement record.

7. Events that produce useful work

Start with one organiser and a three-event sequence. Choose dates once the hosts and leads commit; this document does not create bookings or recurring calendar tasks. Keep a weekly technical session between the larger events where capacity permits.

Event	Proposed hosts/contributors	Required output
Scenario and recruitment workshop	JCI, Startschool, operational contacts and Ukrainian participants who opt in	Agreed problems, a small set of contributor tasks, named reviewers and a participant list
Build and measurement session	Make Riga, radio community, senior engineers and RTU students	Working components, pinned configurations and recorded test results
Controlled outage exercise	Institutional/municipal partner, JCI volunteers and technical team	End-to-end observations, failures, user feedback and a decision about the next stage

An event should yield reviewed code, measurement logs, a test-site commitment or a useful sponsor introduction. Demonstrations should expose limitations and show recovery behaviour. Use synthetic events and requests, clearly distinguished from real alerts. Radio experiments use permitted settings and agreed facilities; interference/abuse testing uses simulation or an appropriately controlled authorised setup.

After the first cycle, assess repeat attendance, maintainer capacity and the value of the results before extending the programme. Give volunteers a path from a small supervised task to equipment care, test coordination or a continuing development role.

8. Budget and funding strategy

The following EUR 30,000 target is an illustrative cash plan, excluding VAT and Kristaps's own time. It assumes access to borrowed phones/laptops and existing premises. It fits within the earlier demonstrator range but adds an explicit independent-review and contingency allowance. Obtain quotations and availability commitments before treating it as a funded delivery schedule.

Use	Planning allowance
Radios, gateways, accessories and spares	EUR 2,000
Software and radio engineering	EUR 18,000
Focused independent security review	EUR 3,000
Field exercises, travel and test logistics	EUR 2,000
Contingency	EUR 5,000
Total	EUR 30,000

Allocate approximately EUR 5,000 first for a bounded feasibility stage; it is included in the total. Commit the remaining approximately EUR 25,000 only against a reviewed result and a credible implementation plan. Donated labour reduces cash spending, not the underlying work. A focused review at this budget is not certification of an emergency service.

Seek one anchor sponsor or two smaller sponsors for the first EUR 5,000, using JCI/business contacts and the Austin partner's introductions. Later sponsor requests can be EUR 5,000-10,000 against a named milestone. Combine cash with documented loans of equipment, premises and engineering time. Offer participation in demonstrations and agreed research outputs; keep operational decisions and resident data under project governance.

The broader engineering model in Appendix A assumes EUR 400-700 per engineering day and 25-50 engineering days for a minimum end-to-end demonstrator. The EUR 18,000 labour allowance is roughly 26-45 days at those rates. The work plan must be resized if quotations or required expertise exceed that allowance. A public municipal deployment has a substantially larger budget and operating burden.

Funding opportunities checked on 8 September 2026. These are candidate routes, not awards or a confirmed combined funding package. Confirm current availability, applicant eligibility, suppliers, cost dates, co-financing and permitted combinations before committing expenditure. Keep each claimed cost allocated once; a voucher is payment for eligible work rather than unrestricted operating cash.

Route	Verified programme information	Proposed next action
LIAA innovation vouchers ^[3]	Applications open while funding remains. Introductory voucher up to EUR 10,000 at 100%; classic voucher up to EUR 25,000 at 85%, subject to conditions. Most supported work uses a qualifying external provider.	Request eligibility advice and a provider quotation for a defined feasibility, experimental-development or testing assignment
LIAA business incubation ^[4]	Intake 9-24 September 2026. Qualifying companies up to five years old need an MVP; listed support includes 70% co-funding and additional prototyping support for innovative companies.	Ask LIAA to assess the applicant company and existing app; prepare an application if eligible
Horizon Europe Cluster 3: disaster resilience ^[5]	2026 call deadline 5 November; decisions anticipated April 2027 and grant agreements July 2027. Topic-specific fit is required.	Ask a research partner or National Contact Point for a consortium already addressing a suitable topic; use later-stage R&D as the proposed contribution
Latvian Defence Ministry grants ^[6]	2026 deadline 18 September; information webinar 11 September. Priorities focus on autonomous military platforms and related capabilities.	Obtain a brief fit assessment before preparing an application; the civilian app alone is not an obvious match

Inga can help reach the appropriate programme owners and potential partners. Andrejs may help sharpen a technical contribution or introduce a sponsor. The Austin partner can pursue cash sponsorship and research contacts in parallel. Record pending applications separately from committed cash and plan the immediate stage around funds actually available.

9. The 12-week sequence and spending gates

The delivery clock starts after the team, initial funds and test access are committed. The following sequence assumes a small team with sufficient combined engineering time; it is not a promise based solely on volunteer availability.

Timing	Required result	Proposed decision
Before start	Two-page brief, named integration/radio leads, candidate institutional counterpart and site, costed feasibility assignment	Approve the initial allocation and responsibilities
Weeks 1-2	Offline cold start, signed update on borrowed phones, initial radio transfer and documented failures	Review feasibility; resize or stop unsupported parts before committing remaining funds
Weeks 3-6	Complete radio/gateway/phone/nearby-phone path, durable encrypted report and delayed return receipt	Confirm that the shared protocol and implementation work together
Weeks 7-9	Repeated field/device tests, congestion, power interruptions, stale or malicious messages and operational walkthrough	Resolve critical findings; record service limitations and excluded conditions
Weeks 10-12	Observed controlled exercise, reproducible evidence, cost model and next-stage proposal	Decide whether to pursue a larger municipal pilot

Agree thresholds before testing. Appendix A proposes at least 95% of compact alerts within two minutes and 99% within five minutes on declared surveyed routes with awake apps under specified load; at least 95% successful foreground nearby exchanges in 30 seconds after setup; durable queued reports; and no accepted forged official messages or false centre receipts in the test suite.

Use at least 100 repeated opportunities per important link/state and report sample counts and uncertainty. These are initial engineering gates, not a demonstrated safety-critical service guarantee. Count device-specific failures and excluded conditions. If background behaviour fails the requirement, retain explicit foreground operation and dedicated gateways as the supported service model.

The reviewer should see both the successful round trip and failures: a removed node, an old event revision, a forged receipt, a restart with queued data and a delayed route back to the server. The institutional observer should assess wording, trust, user comprehension and operator workload as well as technical delivery.

10. Decisions needed before a larger rollout

Use the first pilot to identify an operator, a buyer and a recurring budget. Discuss installation, training, battery replacement, software updates, signing-key operations, exercises, support and who reviews incoming reports. A possible commercial offer is installation/integration plus annual maintenance and exercises, while citizen access remains free. This is a proposition to validate through buyer conversations.

The earlier municipal scenario of roughly 20 service points and 100-300 participants has an indicative EUR 75,000-225,000 scope budget. The wider illustration of 500 local points and 20 regional hubs is EUR 0.72-1.70 million initially with an EUR 100,000-300,000 annual operations placeholder. These are analytical examples, not quotations, funded plans or blanket national coverage. Stage figures should not be added without accounting for reused work and equipment.

Prioritise existing premises, power, staff and distribution arrangements. Independent regional connectivity remains necessary for fresh national information. Local radio nodes and volunteers do not substitute for an agreed upstream path, content authority or an organisation operating the service.

An Austrian evaluation or European consortium can follow once the Latvian pilot produces credible evidence. Seek equity when there is a defensible product, adoption evidence and a purchasing route. Keep subsequent applications anchored to the capability actually measured.

11. First two weeks: action register

All owners below are proposed. Kristaps should confirm people, dates and availability before marking any item committed.

Action	Proposed owner / route	Concrete output
Prepare the introduction package	Kristaps, reviewed by mentor	Two-page pilot brief with architecture, three scenarios, funding target and one tailored ask per recipient
Review architecture and staffing	Andrejs plus senior technical contacts	Short review, named integration candidate and implementation estimate
Confirm radio lead and scope	Dzintars, name to confirm	Four-week assignment, time estimate, equipment loan list and initial test plan
Agree operational requirements	Kristaps with IDC, VUGD and crisis-centre contacts	Named counterpart, reviewed scenario and candidate municipal site
Recruit supervised contributors	Startschool, RTU and Make Riga	A small team, one-to-three-day starter tasks, named reviewers and first test-session date
Assign the event organiser	JCI and Kristaps	Owner and draft plan for the three-event sequence
Gather experience and usability findings	Ukraine-support contacts	Four to six voluntary interviews and a short list of design implications
Seek the first funding commitment	Kristaps, JCI contacts and Austin partner	A defined sponsor proposal for the initial EUR 5,000 stage; record actual commitments
Check grant eligibility and introductions	Inga / relevant LIAA programme owner	Eligibility response, suitable provider route and an application decision before relevant deadlines
Establish international work plan	Austin partner and Inga	Separate introduction targets, sponsor/buyer questions and a 30-day review date

12. Short outreach drafts

These are suggested messages for adaptation. They have not been sent, and none should imply that an institution or named individual has already committed to participate.

Institutional counterpart: We are preparing a small Atbalsts pilot to test trusted information delivery during a prolonged cellular outage. Could we review three concrete scenarios with a nominated practitioner and agree what an exercise would need to demonstrate? We would bring a proposed architecture, test plan and budget, and document the limitations alongside the results.

Mentor / senior adviser: We have an architecture proposal and institutional questions to validate. Could you review a two-page pilot brief and help us reach a senior implementation engineer, a municipal pilot owner and a potential sponsor for the first EUR 5,000 feasibility stage? Each contribution would have a defined scope and reviewable result.

Andrejs: Could you give us a 45-minute review of the Atbalsts architecture and pilot plan, and suggest an experienced engineer who could assess the implementation? A possible contribution around approved multilingual crisis instructions and terminology would also be useful to explore.

Dzintars, once identity is confirmed: Would you help lead a four-week radio feasibility phase: review the design, recommend a small equipment set, mentor a couple of students and run one controlled field test? We would agree your time, available equipment and any paid work upfront.

Inga: We are preparing a Latvian demonstrator with a defined budget and measurable tests. Could you introduce us to the appropriate LIAA funding adviser and one Austrian applied-research or civil-protection contact? We can provide a concise brief showing the actual status of our prospective partnerships.

Sponsor: We are seeking support for a bounded feasibility stage in a civilian crisis-communications pilot. Would you consider funding a defined milestone, lending equipment or contributing engineering time? We would agree the deliverables, demonstration and reporting before work begins.

Contributor: We have small practical tasks in offline data, phone testing, radio logging and gateway tooling. Each task has a reviewer, fictional test data and a demonstrable result. The first assignment is designed to fit roughly one to three working days, with a path to a continuing role if the collaboration works well.

Appendix A. Technical architecture and security

Complete technical assessment from 8 September 2026. Findings are tied to the inspected code revisions; estimates and tests are identified as proposed. The later coordination plan adds the named network and a staged funding target.

Recommendation: build an application that works from local data, exchanges independently verifiable messages over several transports, and uses powered community gateways for dependable relaying. Use LoRa for small messages within surveyed local areas; use local Wi-Fi and Bluetooth for the connection to phones. Test audible data over existing radio broadcasts as an additional downlink. Expand through regional distribution and known service points.

This is a proposed design grounded in the supplied presentation, the available app code, and primary technical sources. No Atbalsts radio, acoustic, range, battery, or phone-background performance has been measured in this review. Prices, capacities, storage sizes and acceptance thresholds below are planning assumptions unless explicitly identified as observed.

A1. What the current evidence supports

The supplied presentation's slide 6 describes approximately 10-30 bit/s, P2P forwarding, and a mobile data node. It does not identify a modulation, frequency, radio receiver, tested protocol or measured throughput. Slide 8 explicitly attributes video publication to ePPO. These are claims to clarify; the presentation is source material, not an implementation specification.

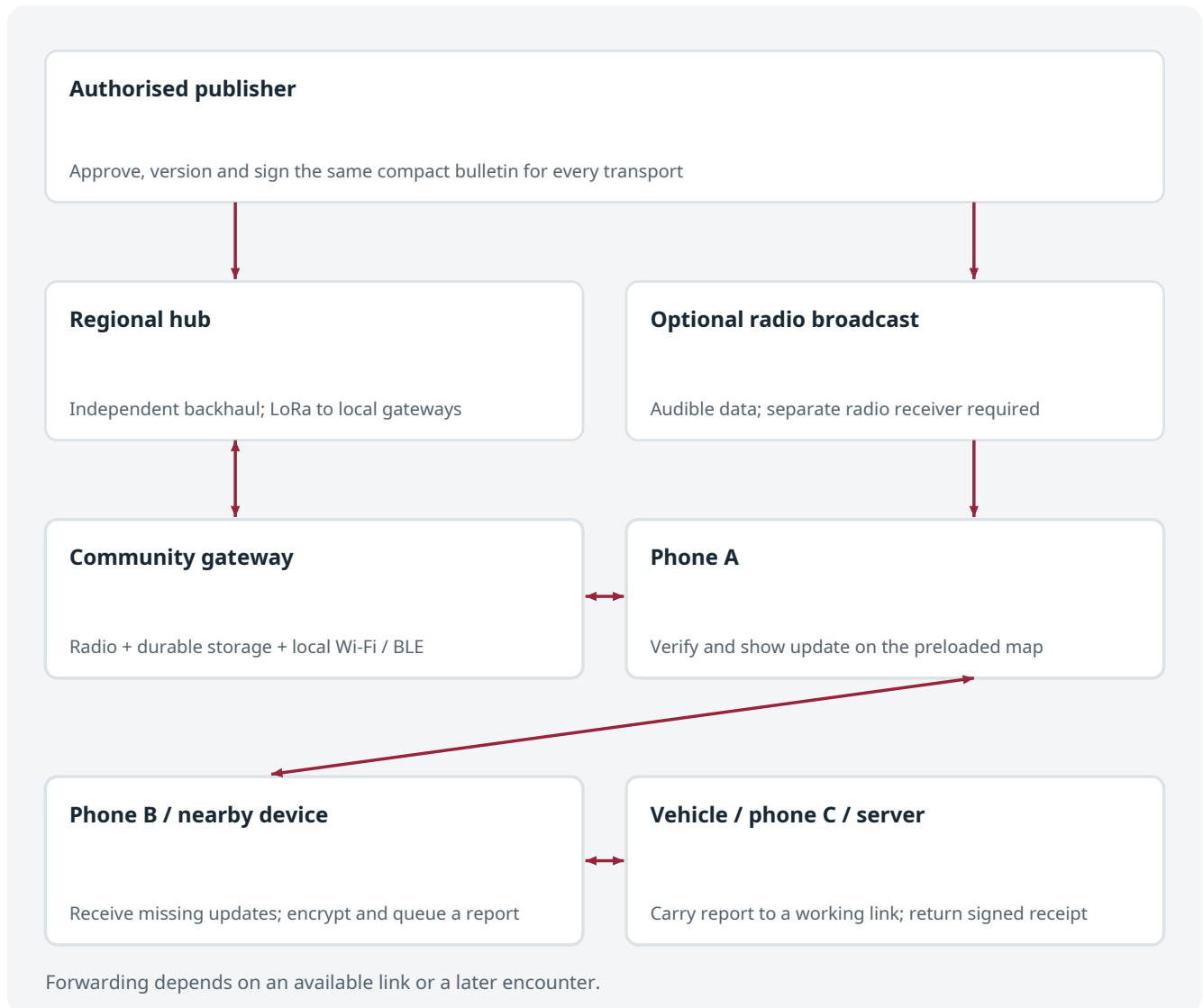
I checked GitHub's default branch, `goofis11/palidzi`, at `b7ee07c7e57801cbab1202edbef7b0b798bfcc89`, and the local working branch `fix/r2-atomic-publication` at `60a88a7`. The repository is private. This verifies those code revisions, not the exact revision running at `atbalsts.sortium.co`.

Observed in the code	Architectural implication
A React/TanStack web application with a service worker caching the public /72h page, assets and eligible reference geodata	A useful preparedness foundation; the map and live API responses are not a complete offline application
Leaflet basemaps request external tiles/WMS	Add an explicitly packaged offline basemap and local renderer/data source
The report form posts to /api/reports; local storage keeps receipts after successful submission	Add durable local report storage before attempting transmission
No LoRa, Bluetooth transport, native mobile project, or signed offline bulletin implementation found in the reviewed application files	These are new components, not configuration switches
A server publication outbox exists in the newer local branch; it is absent from the checked GitHub main tree	Reuse the outbox pattern, but verify its release status and implement transport delivery separately

Code references: [GitHub service worker at the checked revision](#)^[7], local basemap configuration (`lib/basemaps.ts:103`), report submission (`routes/zinot.tsx:135`), and local server outbox (`lib/outbox.ts:1`).

The two local public GeoJSON files total **1,345,977 bytes**, approximately 1.35 MB, uncompressed. They contain municipality boundaries and shelter reference data. That is not the size of a complete offline Latvia map, nor evidence that every necessary resource is cached.

A2. The recommended system



There are three different connectivity problems:

1. **Reaching an isolated region.** A regional hub needs a path from an authorised publisher: surviving fixed internet, an agreed government link, satellite, an engineered radio link, a broadcast receiver, or physical transport. LoRa does not create this upstream path automatically.
2. **Distributing within the region.** A modest number of elevated, powered LoRa nodes exchange compact updates. Public access points cache them and make them available over local Wi-Fi/BLE.
3. **Reaching people away from those points.** Phones carry verified updates and exchange them during encounters. This helps, but arrival depends on people, permissions, battery and app availability.

An isolated municipality also needs the ability to originate authorised local information. Provision a municipal publishing laptop with a hardware-protected, geographically restricted delegated signing credential and a local interface. It must work without cloud login or the central database. Its allowed message types, operating period and approval procedure should be defined beforehand.

Every transport carries the same signed public objects or encrypted private report objects. The app's verification and storage rules do not change when a message arrives through radio, a neighbour, a USB file or the internet.

A3. What receives the signal, and which technology to use

Link	Physical receiver and phone connection	Appropriate role
LoRa in a permitted European SRD band	Separate LoRa radio; a personal accessory can connect by BLE. A community gateway should attach the radio by USB/serial to a local host and serve phones over Wi-Fi/BLE	Compact alerts, resource status, small requests and receipts
Local Bluetooth LE	Phone Bluetooth hardware, through native APIs	Discovery and small nearby exchanges
Local Wi-Fi	Phone Wi-Fi, using a gateway access point or compatible peer connection	Fast bulletin sync, larger data packages and queued attachments
Audible data carried in FM/AM/shortwave audio	A radio receiver first receives the RF signal; its speaker feeds the phone microphone, or its audio output feeds a gateway	Optional broadcast downlink
Licensed VHF/UHF or TETRA integration	Approved radio terminal/gateway and operator-supported interface	Potential institutional backhaul where access and continuity are agreed
Satellite	A supported terminal at selected hubs	Independent backhaul; requires equipment, service and power

Ordinary supported iPhones and Android phones should be assumed to have **no usable LoRa receiver**. The microphone hears decoded sound from a separate radio; it does not receive the broadcast RF carrier. Built-in FM access on some Android models is too inconsistent to be a baseline requirement.

Use **Meshtastic-compatible EU radios for the small pilot**, carrying application-signed payloads through the device API. Pin the firmware/configuration and test packet size limits. Avoid assuming the stock chat application, its cache, or its optional forwarding modules implement Atbalsts delivery semantics. Meshtastic provides [serial/TCP/BLE client interfaces](#) ^[8].

For the first deployment, keep the radio topology small and managed, preferably one hop with a few deliberate relays. Compare faster modem settings against maximum range. A broadcast flooding mesh becomes inefficient as retransmissions multiply; adding devices does not necessarily add capacity. Meshtastic documents its [broadcast algorithm](#) ^[9].

LoRa and LoRaWAN are different layers. LoRaWAN normally uses gateways and a network server, with gateway backhaul over IP. A surviving public LoRaWAN gateway does not imply that the application still works when its server/backhaul is unreachable. An existing network could be useful if local server operation, power, downlink scheduling and failure behaviour are explicitly supported. [Semtech architecture](#) ^[10].

A4. The “10-30 bit/s” claim and a sensible audio experiment

The slide's number should be described as an **unverified design assumption**. It is not the identity of a radio technology.

For a concrete audio candidate, start with **audible MFSK16 modem tones centred within the receiver's normal audio passband**, initially around 1.5 kHz. MFSK16 carries approximately 31.25 information bits/s after its modem

error correction, before application framing and text/binary encoding costs. That makes an eventual 10-30 useful bit/s hypothesis plausible, but does not establish it for this application. Test faster MFSK profiles if the broadcast path supports them. [Modem developer's explanation](#) ^[11], [Fldigi mode table](#) ^[12].

The practical sequence is:

1. The publishing service produces a compact signed bulletin.
2. An audio encoder adds framing, error correction and an identifiable start sequence.
3. An authorised broadcaster inserts a short audible data burst between spoken messages.
4. A listener opens "Receive radio update" and places the phone near a battery radio.
5. The decoder reconstructs the complete object. Only successful signature verification permits an official update on the map.
6. The phone can later share the original bytes with neighbours.

This approach has precedents: [TIVAR's project documentation](#) ^[13] describes Android decoding through a microphone or audio cable; [Shortwave Radiogram](#) ^[14] describes actual data broadcasts. Those demonstrate components, not a maintained, audited Atbalsts implementation for current iPhones.

Do not make near-ultrasonic sound the broadcast baseline. Standard FM programme audio occupies roughly 0-15 kHz, so signals above that range cannot be assumed to survive the broadcast chain. Cheap receivers, speaker response, room noise, processing and phone microphones add further uncertainty. This is distinct from the radio-engineering term HF, which refers to a radio-frequency band. [ITU FM baseband description](#) ^[15].

[ggwave](#) ^[16] is useful for a quick audible speaker-to-phone experiment. Its documented audible configuration spans roughly 4.5 kHz; it is not automatically suitable for a narrow voice-radio audio channel. Near-ultrasonic phone-to-phone experiments can remain optional; they add little reliability compared with BLE, Wi-Fi or a signed QR exchange.

RDS or another dedicated broadcast data service is a possible later way to avoid audible bursts, but requires compatible data receivers and broadcaster integration. A normal radio speaker does not expose an RDS data stream to the app.

Broadcast reception is **one-way**. A phone cannot reply to the FM transmitter through its microphone. Replies must use a gateway/P2P path, surviving connectivity, or physical carriage.

Broadcast airtime also needs a schedule and a budget. At a delivered 30 bit/s, ten separate 160-byte regional bulletins take more than seven minutes before repetitions. Publish short common alerts broadly, group changes where appropriate, schedule regional content and let gateways cache broadcasts continuously. A household listening briefly can miss the beginning or its region's update. Measure that user experience, not just decode success for an already-running receiver.

A5. Make the messages small without removing their meaning or security

Use an authority-facing alert model compatible with CAP concepts, then define a compact radio profile. CAP provides useful update, cancellation, geographic and multilingual semantics; full XML need not cross LoRa. A custom compact profile is not automatically CAP-conformant. [OASIS CAP 1.2](#) ^[17].

Proposed compact public bulletin fields:

- Protocol version, issuer/key identifier and an explicit exercise/real-message distinction.
- Event identifier, revision, issue time, effective/expiry times and cancellation/update relationship.
- Hazard, severity, instruction code and relevant parameters.
- Coordinates with a circle/radius, a predefined area identifier, or a small explicitly transmitted geometry.

- Codebook/package version, optional short text, and digital signature.

Target approximately **120-200 bytes for a simple signed bulletin**, subject to real serialization tests. More complex geometry, long text, delegation certificates or multiple signatures need more bytes or additional packets. The budgets must count all of them.

Use a standard signed-object format such as COSE Sign1 with a supported signature algorithm, for example Ed25519. An Ed25519 signature alone is 64 bytes. Preinstall the public trust material; relay devices must not possess authority signing keys. [COSE specification](#) ^[18], [Ed25519 specification](#) ^[19].

The app can expand “water distribution point; object 241; open until 18:00” using an approved local dictionary. Keep dictionaries versioned and signed: silently interpreting an old instruction code through a different dictionary is a safety failure. Include enough basic hazard/geometry information for a safe generic display when an optional package is missing. Never infer a detailed instruction from an unknown code.

Assuming the stated rates represent **delivered application bytes**, the arithmetic is:

Complete data object	At 30 bit/s	At 10 bit/s
160-byte signed bulletin	43 seconds	128 seconds
1,000 bytes	4.4 minutes	13.3 minutes
100 kB image	7.4 hours	22.2 hours
1 MB	3.1 days	9.3 days

These are lower-bound transfer times, excluding any costs not already included in the measured rate. Actual audio framing, text encoding, missed beginnings and repetitions increase latency. A raw modem bit rate must not be presented as delivered application throughput.

Do not send photos or video through the narrow radio channel. Preloaded icons and explanatory illustrations are fine. A new photo or video can remain encrypted on the originating phone, travel over local Wi-Fi to a vehicle/gateway, and upload when capacity returns. Send its small report summary first. The UI must distinguish illustration, locally stored attachment and attachment actually received by the centre.

A6. Phone-to-phone exchange: feasible, with an explicit operating model

Build a native iOS/Android communications layer while reusing suitable React interface and business logic. Bundle the app shell and essential data locally; a wrapper that loads the website remotely does not solve offline startup.

A public gateway should expose a small local sync service, not require the full cloud backend. Joining its hotspot must be an explicit, tested flow, including the phone's “no internet” behaviour. Discover it through declared local-network services, BLE or a QR containing connection information. iOS requires appropriate local-network permission declarations. [Apple local-network documentation](#) ^[20].

Do not assume a hotspot can transparently replace the existing HTTPS website. A page at a local address has a different origin and cannot simply write the existing website's local database. Browser storage is also subject to eviction. Use a preinstalled app that imports verified objects into its own durable storage. This recommendation follows from [WebKit's origin-isolated storage](#) ^[21] and [storage policy](#) ^[22].

Provision distinct gateway identity credentials and an offline-verifiable trust chain for authenticated local sessions; test credential validity across the intended outage. Never distribute the main website's TLS private key to all gateways or teach users to bypass certificate warnings. Public objects still need their authority signatures, even over authenticated Wi-Fi. A portal downloaded from an unknown hotspot cannot establish its own authenticity

merely by displaying a “signature verified” badge; a previously installed verifier or a trusted staffed service point is needed.

The baseline user action should be **“Exchange nearby updates” with the app open**, with clear consent and controls for relaying private report envelopes. Advertise a dedicated service, connect, exchange short inventories, transfer missing objects, verify and commit locally. Use BLE for small exchanges and a gateway Wi-Fi network for larger ones. Keep a signed QR/file-transfer fallback for a few important bulletins when pairing fails.

Google's **Nearby Connections** ^[23] is a candidate for native offline peer exchange; its SDK handles several underlying transports. It does not provide an automatic delay-tolerant nationwide network, or remove the need for application-level authentication.

Current Apple support matters: **iOS 26 introduced Wi-Fi Aware on iPhone 12 and later**. This is a useful enhancement for compatible devices, not a reason to exclude older phones. Apple's API includes pairing and service-discovery flows; evaluate those interactions before promising frictionless sharing with strangers. **Apple's Wi-Fi API overview** ^[24], **Wi-Fi Aware presentation** ^[25].

Background behaviour must be described honestly:

Situation	Product commitment
Both apps open, permissions granted, radios enabled	Primary supported exchange mode; test each platform pairing
iPhone communicating with an associated BLE accessory	Some background events are supported; not a continuous-execution guarantee
Arbitrary iPhone-to-iPhone encounters with apps suspended	Opportunistic only; no delivery guarantee
Apple Multipeer Connectivity app enters background	Its documented advertising/browsing stops and sessions disconnect
Android background service	Potentially more capable, but subject to permissions, foreground-service rules and device power management
App force-quit, device restarted, battery saving or radios disabled	Explicit test cases; never presume relaying continues

Sources: **Apple Core Bluetooth background rules** ^[26], **Multipeer Connectivity** ^[27], **Android background BLE** ^[28].

Initial **test distances**, not guarantees: 5-20 m for phone BLE encounters with people/walls present; 20-50 m for a local Wi-Fi access area. Good outdoor conditions may allow more. Building materials, crowds, device orientation and interference can substantially reduce them.

Always-on relaying belongs primarily on purpose-built gateways and supervised vehicle devices, where execution and power are under operator control.

A7. Store-and-forward and returning citizen reports

Use a durable database on each phone and gateway. Persist an outbound report before starting any network operation; preserve it across termination and reboot. Track public bulletins, private report envelopes, attachments and receipts separately.

Store-and-forward is an established networking approach; **Bundle Protocol v7** ^[29] is a reference for delay-tolerant semantics. Evaluate an existing implementation for hub/vehicle interoperability, but measure its complete

overhead before choosing it for tiny radio packets. The application still needs clear event, trust and receipt semantics even if a standard transport is reused.

For public information, exchange an inventory of active event IDs and revisions and request missing versions. Use periodic signed regional manifests/snapshots to detect gaps. A single global “highest sequence seen” is insufficient: receiving a later update must not hide a missed update to a different event.

Deduplication should use a content hash of the immutable signed object plus issuer/event/revision semantics. Verify and render one event version once. Superseded versions and cancellation tombstones must survive long enough to stop old devices resurrecting an alert. Include bounded storage, per-peer quotas, randomised retry/backoff and a forwarding budget. Hop limits help normal operation but cannot stop a malicious radio from ignoring them.

Maintain a hierarchy of queues: trust/revocation information and official urgent updates first; important small citizen reports and server receipts next; routine status later; media and full packages only on larger links. Both directions require reserved capacity. A downlink-only success demonstration is incomplete.

Citizen reports should be encrypted on the originating phone to a centre/report-ingest public key using a reviewed standard construction such as **HPKE**^[30]. Relay phones can carry the ciphertext without seeing the person's location, contact details or medical information. Retain metadata minimisation and traffic quotas because encrypted content can still be spam. Keep signing and report-decryption keys separate.

An account should not be required to read official warnings. If unregistered offline reports are permitted, label their identity assurance and route them to verification; do not invent an online identity check. A citizen signature proves possession of a key, not that a sighting or request is true.

Return distinct states:

Saved on this phone -> copied to a relay -> received by the centre -> reviewed/acted on.

A relay receipt is not a server receipt. A server receipt is not a promise of dispatch. Sign centre receipts and route them back using the same store-and-forward mechanism. Delivery time is unbounded while no path or physical carrier reaches the server.

A8. Offline maps, new objects and several days without internet

Ship a small survival package with the installed application: verified instructions, trust keys, message dictionaries, an overview map and essential reference points. Offer a separately verified detailed Latvia map download before a crisis, with visible size, date and completeness. Installation, key provisioning and permission setup should happen before the outage; do not assume an unprepared iPhone can install the app from a neighbour. Staffed information points and ordinary spoken radio remain necessary for people without the installed app.

Proposed storage targets, to measure during the pilot:

Package	Planning allowance
App, essential instructions, fonts/icons, overview map and core reference points	50-100 MB
Detailed, selected-layer vector map of Latvia	Additional 200-800 MB
Bulletin history and reserved private report storage	20-100 MB
Photos/video	Separate user-controlled quota

These are design allowances, not measured outputs or a claim about existing app size. Full imagery, more zoom levels, routing graphs and multilingual media can increase them substantially.

Store map tiles, fonts, styles and sprites locally. Use licensed offline map data or generate packages from permitted source data. **Do not bulk-download the current public OpenStreetMap tile endpoint:** its policy prohibits offline packages. [OSMF tile policy](#) ^[31].

During an outage, preserve the basemap and update a separate signed event/resource layer. Radio can carry a new water point with coordinates, short label, status and validity, or a circle/small polygon indicating a closure. Full road-network revisions and detailed imagery wait for Wi-Fi, physical delivery or restored internet.

Make ordinary alert updates self-contained wherever possible; a user should not need every preceding revision. For genuinely differential packages, include the required base version and missing-chunk recovery. Apply changes atomically after all checks pass.

Show both the reference package date and each event's last verified issue time. **Expiry means that current status is unknown; it does not mean the danger has ended.** Do not automatically show a shelter as open or remove a hazard merely because no fresh data arrived.

Location can work from satellite positioning without cellular data, but indoor reception, cold starts, jamming and spoofing remain concerns. Provide manual map placement and label location accuracy. The offline assistant should retrieve approved local instructions deterministically; a cloud-based assistant or a server-side fallback does not automatically run on the phone.

A9. Coverage, radio capacity and power

For budgeting, start with **0.5-2 km in an obstructed urban setting and 3-10 km in rural conditions with reasonable antenna placement**, between LoRa radios. Elevated, clear paths can exceed these figures; basements and body-worn radios can perform much worse. These conservative starting assumptions need site surveys. Semtech's own general examples describe up to about 5 km urban and 15 km rural, which are not service guarantees. [Semtech range discussion](#) ^[32].

LoRa coverage to a gateway is not direct coverage to ordinary phones. The gateway's local Wi-Fi/BLE area is much smaller. A person outside it needs an accessory or an encounter with another informed device.

Latvia is approximately 64,600 km². Even ideal hexagonal coverage gives about 995 sites at a 5 km radius, 249 at 10 km and 111 at 15 km. The calculation is area divided by $2.598 \times \text{radius}^2$. It ignores terrain, borders, overlap, indoor coverage, redundant routes, power and capacity; it is not a deployment plan. [Latvian statistical area reference](#) ^[33].

Start with coverage of **named places and populations**: shelters, municipal offices, libraries, medical/resource points, and planned vehicle stops. Report accessible service locations separately from territorial RF coverage and actual people reached.

Latvia's frequency plan provides, for nonspecific SRDs in 869.4-869.65 MHz, up to 500 mW ERP and applicable spectrum-access/mitigation requirements, with a duty-cycle alternative of no more than 10%. Other subbands have different conditions. Use compliant equipment, antennas and settings; do not assume a software power value alone establishes compliant ERP. Confirm the complete deployment configuration with the spectrum authority before field transmission. [National frequency plan, Annex 3, table 8.14, row 54](#) ^[34].

Meshtastic EU868 LongFast is nominally about 1.07 kbit/s and uses one 250 kHz frequency slot in this band. That nominal rate excludes packet and network overhead. A different chat-channel name or encryption key does not create more RF capacity. [Meshtastic radio settings](#) ^[35].

The standard LoRa airtime formula gives approximately **1.68 seconds for a 200-byte radio payload** at SF11, 250 kHz, coding rate 4/5, eight preamble symbols, explicit header and CRC. This is a calculation, not a measured Meshtastic application transfer.

Using a rounded two-second packet as a planning example:

- A transmitter operating at 10% duty cycle has at most 360 transmit seconds/hour: 180 such transmissions.
- If that bottleneck transmitter sends each unique bulletin three times, only 60 unique bulletins/hour remain before other traffic.
- Reserving half of that airtime for other work leaves around 30 such bulletins/hour.
- A 1% duty-cycle budget would be ten times smaller under the same packet-time assumption.

This is not network-wide capacity: contention, hidden transmitters, receiving while another node transmits, retries and relays reduce practical delivery further. Regional targeting, selective repetition, faster settings on good links and geographic frequency reuse matter more than adding indiscriminate repeaters.

One radio broadcast can be heard by many receivers without per-recipient airtime. Thousands of individual acknowledgements or help requests are a different problem. Do not request one radio acknowledgement from every citizen.

For a public service point, use a local host with USB-connected radio and a separate Wi-Fi access point. Treat the radio's stock BLE link as a personal/maintenance interface. Meshtastic also documents that its ESP32 firmware disables Bluetooth when Wi-Fi is enabled; do not assume both public interfaces are simultaneously available on one cheap board. [Bluetooth configuration](#) ^[36].

Set an initial local gateway test target of **20 simultaneous clients and 100 small sync sessions per 10 minutes**. Larger targets, such as 50 concurrent clients, require suitable AP hardware and measurement. Local Wi-Fi clients are not individual LoRa senders; service capacity is limited separately by local sessions, RF airtime, storage and return-link availability.

Power must be budgeted from measured watts. A gateway averaging 3 W needs about **270 Wh of nominal battery capacity for 72 hours** at 80% usable energy. A nominal 20,000 mAh, 3.7 V power bank supplies only about 20 hours at that load. Winter solar output, battery temperature, charging losses and Wi-Fi duty cycle require explicit tests. Low-power radio nodes and continuously running Wi-Fi hosts have very different power profiles.

A10. Security: contain failures instead of trusting the network

Public bulletins should be readable but authenticated. Private reports should be confidential to the centre. These are separate requirements; “the mesh is encrypted” does not establish either one by itself.

Threat	Required design response	Residual limit
Forged warning, changed coordinates or fake all-clear	Verify the authority signature on the complete object before accepting it; bind issuer, content, times, scope and revision	A compromised authorised signer can still sign false content
Compromised phone or community gateway	Keep authority private keys elsewhere; reverify received objects on every device; separate maintenance and public interfaces	It can lie on its own display, censor, delay or flood
Old valid alerts replayed after cancellation	Persist revisions and cancellation records; exchange signed manifests and freshness state	An isolated device cannot know about a cancellation it has never received
Stolen publishing credential	Hardware-protected keys, restricted regional/type delegation enforced by every receiving app, operator authentication, approval controls, audit and exercised recovery	Instant revocation cannot reach a fully isolated device

Threat	Required design response	Residual limit
Fake hotspot or malicious peer	Verify objects independently; use authenticated transport where available; end-to-end encrypt reports before leaving the phone	Traffic timing and presence may still be observable
Flooding and fake citizen identities	Bound frame sizes, parser work and storage; impose per-peer/traffic-class quotas; reserve official-message capacity	An open offline reporting network cannot eliminate Sybil spam
Radio jamming, failed mast, power loss	Diverse paths, separated sites, measured backup power, physical carriage and visible stale-data states	Software cannot guarantee availability under sustained physical interference
Clock manipulation or GNSS spoofing	Track last trusted time and monotonic elapsed time where available; resist rollback; expose uncertainty; allow manual location	After long isolation/reboot, exact freshness may be unknowable
Parser exploit or malicious map package	Minimal bounded binary schema, strict validation, signed manifests, fuzzing and independent review	Device/OS compromise remains possible
Malicious build/update	Signed releases, protected build credentials, dependency review, controlled rollout and rollback/recovery plan	Protecting the signer and release chain remains essential

Use a long-lived root trust anchor protected offline and a manageable set of scoped operational credentials. Distribute current/next credentials and signed revocation information before predictable rotation boundaries. Define an offline validity period aligned with the intended outage duration. Never silently bypass expiry or signature checks to recover availability; retain access to static preparedness information and label unverifiable updates.

Administrative revocation and report-decryption key rotation must preserve the ability to process legitimate delayed reports. Define retention and receipt expiry rules, avoid broadcasting identifying report metadata, and keep location telemetry disabled unless required for the specific role.

Meshtastic's default public channel uses a publicly known key; its newer private-message security does not make a radio node an authorised national publisher. Put the authority verification above whichever radio system is selected. [Meshtastic encryption documentation](#) ^[37].

For nationwide or consequential messages, require strong publication controls and independent approval appropriate to the urgency. Maintain a documented urgent-issue procedure, especially for locally delegated operators. Cryptography verifies who issued a message, not whether the operator's underlying facts are correct.

A11. A realistic end-to-end scenario

Example: a storm has disabled cellular coverage in a municipality. Its local gateway is powered and still has a surveyed LoRa path to the regional hub. At least that regional hub has independent connectivity, or receives messages through the agreed broadcast/physical route.

1. An authorised operator confirms that a resource point is closed and an alternative is open. The publisher creates event revision 7, with validity, coordinates and instructions, and signs it. The publication and delivery work are stored durably.
2. The regional hub obtains the message and sends it into the local LoRa network according to airtime and priority limits.

3. The community gateway receives, verifies and stores it. Phone A joins the local Wi-Fi service or uses a tested BLE connection. The app verifies the same signed object, applies it to its offline map and shows source and age.
4. Phone A later meets phone B, which never received radio or visited a gateway. Both users open nearby exchange. B requests the missing event revision and receives the original signed bytes.
5. B creates a help request. Its location and details are encrypted locally; the app initially says it is saved on the phone.
6. B copies the request to A or a vehicle gateway. A small envelope may travel upstream by LoRa if capacity and a connected route exist. Otherwise it stays queued until a vehicle reaches a working hub.
7. The centre receives the report, deduplicates it and issues a signed receipt. Human review and operational action are tracked separately.
8. That receipt travels back through available gateways/peers. Until B receives it, the app does not claim that the centre received the request.
9. If the official situation changes, revision 8 or an explicit cancellation propagates in the same way. Devices that missed it show their last verified information and its age.

An awake phone near an uncongested, connected gateway could receive the initial bulletin within minutes. A phone depending on a future encounter might wait hours or never receive it. These are different service classes, not one end-to-end guarantee.

This distinction matters for the presentation's air-threat example: encounter-based forwarding cannot carry a guaranteed seconds-level warning. Use the proposed system first for outage information, resource availability and delayed reports; any time-critical warning role needs its own tested delivery requirement alongside the established official warning channels. Do not derive a safe evacuation route from stale closures or an offline model's guess.

A backpack/vehicle node can combine receiver, retransmitter, cache, local hotspot and data carrier. The vehicle must stop long enough for discovery and transfer. Start with vehicles and fixed sites; a drone adds endurance, radio geometry, payload, aviation and operational constraints without fixing the application's trust or background-execution problems.

A12. Budget options

All figures below are **planning estimates in EUR, excluding VAT**. They assume borrowed phones/laptops and existing premises where indicated. Labour is valued at an assumed €400-€700 per engineering day; this is a modelling rate, not a Latvian supplier quotation. Donated labour reduces cash spending, not the amount of work.

Hardware price anchors checked during research: RAK's **Meshtastic starter kit** ^[38] starts at US\$24.99, and Raspberry Pi advertises its **Zero 2 W** ^[39] at US\$15. These are bare-product starting prices, not complete European deployment costs. Batteries, cases, storage, suitable antennas, adapters, shipping and mounting materially change the total.

Before buying anything, use borrowed phones and a laptop for a zero-additional-hardware test of signed QR/file transfer, local state, stale updates and report receipts. That can reject a flawed message design cheaply; it does not validate radio reception or background P2P.

Stage	Scope	Hardware/field allowance	Development and other work	Planning total
Bench decision test	Two radios, borrowed laptop/phones, audio feasibility and message format	€300-€600	5-10 days: €2,000-€7,000	€2,300-€7,600

Stage	Scope	Hardware/field allowance	Development and other work	Planning total
Minimum meaningful end-to-end pilot	Six radios, two public gateways, 8-12 borrowed mixed phones; native foreground exchange and return receipts	€1,200-€2,200 including hardware contingency	25-50 days: €10,000-€35,000	Approximately €11,000-€37,000
Controlled municipal pilot	Approximately 20 powered service points, regional/vehicle equipment, 100-300 participating residents	€20,000-€50,000	80-160 days: €32,000-€112,000; independent review/exercises €10,000-€25,000	€75,000-€225,000 including 20% contingency
Illustrative wider rollout	500 local service points and 20 regional hubs on existing sites	See calculation below	Production engineering, independent review, operations setup	€0.72-€1.70 million initially; not blanket national coverage

The bench stage does not deliver the complete native application. The minimum end-to-end stage is a development demonstrator, not a public emergency service. The municipal stage assumes significant reuse of earlier work and targets controlled operation; a procurement requiring formal service guarantees may cost more. Stage totals describe each scope and should not be added without accounting for reused work and equipment.

Minimum pilot bill of materials:

Item	Quantity/basis	Estimate
Complete LoRa kits with suitable antenna, small battery/case	6 × €60-€100	€360-€600
Local hosts, Wi-Fi equipment, storage and adapters	2 × €120-€220	€240-€440
Additional power/testing adapters	Allowance	€200-€350
Audio receivers/cables	Allowance	€60-€140
Mounts, spares and local travel	Allowance	€150-€300
Subtotal		€1,010-€1,830
20% contingency		€202-€366
Equipment/field total		€1,212-€2,196

This minimum does not purchase 72-hour backup for two continuously operating full-power hotspots, paid tower access, production certification, new test phones, vehicles, drones or broadcast airtime. Those are added when moving to a deployment.

The wider-rollout example is deliberately transparent: 500 service points × €500-€1,200; 20 regional hubs × €2,500-€6,000; €100,000-€250,000 for installation/training/logistics; €200,000-€450,000 for production engineering/security/operations setup; then 20% contingency. It assumes usable existing buildings, no new national mast network and mostly reusable backhaul. Allow roughly €100,000-€300,000/year as an initial

operations placeholder for support, maintenance, batteries, connectivity and drills; obtain operator/site quotes before treating it as a budget.

The least expensive scaling path is likely to combine existing broadcast distribution, selected independently connected hubs and local service points. **LVRTC offers FM transmission, planning and coverage services** ^[2]; this is a credible partnership route, not evidence of permission, free airtime or crisis availability for Atbalsts. Existing municipal roofs, power and staff can be more valuable than very cheap radio boards.

Where line of sight and power are available, engineered point-to-point Wi-Fi/microwave can carry far more regional traffic than LoRa. Obtain site-specific quotes and compare it with licensed institutional radio and satellite at selected hubs. Keep LoRa focused on the small-message role.

A13. What to test before increasing the budget

Use gates that can stop or reshape the project:

1. **No-network startup.** Install all required material, disable cellular service and WAN access while leaving local radios enabled, reboot, open the app and display the map/instructions. Repeat after several days. Check missing/evicted/corrupt packages and unavailable login services.
2. **Signed-message round trip.** Centre -> radio -> gateway -> iPhone and Android -> previously uninformed phone -> encrypted report -> delayed server ingestion -> signed receipt. Log the same message ID at every stage.
3. **Phone compatibility.** Test iOS<->iOS, Android<->Android, iOS<->Android and phone<->gateway in foreground, locked, background, low-power, force-quit and reboot states. Include older supported iPhones and multiple Android vendors. Count failures rather than excluding them.
4. **Audio separately.** First prove speaker/cable decoding. Then test the actual intended authorised broadcast and receiver chain, different radios, volume, background noise and phone microphones. Measure complete verified payloads per elapsed minute. Bench playback alone does not prove over-air reception.
5. **RF field survey.** Measure delivery probability, latency, RSSI/SNR, antenna height, obstructions and occupied airtime on urban/rural routes and indoors. Test both directions and at least one relay failure. Never equate the longest successful packet with coverage.
6. **Congestion.** Load local sessions and queues, then test a burst of requests while official alerts continue. Measure queue age and fairness. Simulated clients measure software load; real radios are needed to measure collisions.
7. **Abuse and stale state.** Submit corrupted signatures, altered coordinates, obsolete codebooks, old all-clears, duplicate floods, invalid sizes and forged receipts. Compromise a relay in the test environment and confirm it cannot mint official messages. Test a missing revocation and explain the residual risk.
8. **Power and recovery.** Measure continuous watts, run a 72-hour deployment power test, cut power during writes and recover state. Exercise backup signing, key rotation and an internet return after days offline.

Suggested initial acceptance targets: at least 95% of compact alerts within two minutes and 99% within five minutes **on declared surveyed routes with awake apps, under a specified load**; at least 95% completion for foreground nearby exchanges in 30 seconds after permissions/setup; zero acceptance of forged official messages in the test suite; durable requests across restart; no "received by centre" state without a valid receipt.

Use at least 100 repeated opportunities per important link/state and report sample counts and uncertainty. These pilot targets are not sufficient evidence for a safety-critical SLA. Record failure clusters and device-specific exceptions rather than averaging them away. If iPhone background behaviour misses the required threshold, retain the explicit foreground exchange and gateway model.

Conduct interference/abuse tests through simulation, conducted connections, shielding or an appropriately authorised facility. Do not intentionally jam public radio spectrum.

A14. What to correct before replying to IDC

- Replace slide 6's unconditional radio speed with a named **proposed transport** and measured-result placeholder: "Compact signed alerts through external LoRa gateways; experimental audible radio-data reception; throughput and delivery to be established in a pilot."
- State that external radio hardware is required, except for a microphone receiving sound from an ordinary radio. Explain the short Wi-Fi/BLE connection to the phone.
- Distinguish foreground nearby exchange from opportunistic background operation, particularly on iPhone.
- Describe the map as preloaded reference data plus small signed event/resource updates. Remove any implication that photos/video or full maps travel at 10-30 bit/s.
- Clarify that the offline assistant uses an installed, approved knowledge package.
- Correct the ePPO reference. Its indexed official instructions describe choosing a threat type, providing location and pointing the phone; the current developer listing describes threat reporting. I found no primary evidence supporting the presentation's video-upload or LoRa/P2P architecture claims. Cite it only as an example of citizen observations supporting an institutional response unless the developers provide further evidence. [ePPO official site](#) ^[40], [developer's Google Play listing](#) ^[41]. The official website blocked direct retrieval in this review; its indexed instructions and accessible store listing are a limited evidence base.

There are deployed components: Meshtastic radio messaging, Briar's offline peer synchronisation, and data-over-radio experiments. [Briar describes Bluetooth/Wi-Fi synchronisation](#) ^[42]; it should not be represented as proof of automatic forwarding among arbitrary iPhones. No reviewed source establishes this exact combined Atbalsts architecture at Latvia scale or under sustained hostile interference.

A15. Implementation sequence for the existing app

Keep one public event model and add bounded components:

Component	Work to implement	First proof
Publishing/export service	Turn an approved event revision into an immutable signed compact object; commit durable delivery work with the publication	Repeated delivery cannot change the signed bytes or create a second event
Shared message library	Strict schema, serialization, signature/credential checks, codebook handling, event revision rules and receipt interpretation	The same object yields the same result on iPhone, Android and gateway
Native app shell and local store	Bundle reusable React UI, use durable local storage and add native BLE/Wi-Fi/microphone bridges	Cold start and map display without cloud, DNS, login or remote scripts
Offline map/data pack pipeline	Produce permitted vector packages, manifests, hashes, versioned reference objects and update recovery	A partial/corrupt download never replaces the last complete working package
Gateway daemon	USB radio adapter, local sync server, independent queues, deduplication, priority, quotas and power/restart recovery	An attached radio is replaceable without changing the phone's message format

Component	Work to implement	First proof
Report ingest and receipts	Accept delayed encrypted reports, deduplicate, preserve reported time and identity assurance, issue recipient-verifiable receipts	Many copies of a request create one report; "received" requires server evidence
Operational console	Distinguish publication from each gateway's receipt and from user/device acknowledgements	No inference that every resident received a warning from one gateway's success

The existing server outbox stores event IDs and versions. The radio export must bind delivery to an immutable snapshot of that exact revision; looking up only the mutable latest event during a retry could send different content under the same delivery task. Preserve the current protection against caching private authenticated pages. Build an explicit offline data store instead of broadly relaxing the service-worker allowlist.

Suggested order: (1) protocol and offline startup, (2) gateway-to-phone delivery, (3) delayed reports and receipts, (4) foreground phone-to-phone exchange, (5) independent audio feasibility, then (6) background optimisation and regional deployment. Audio experimentation can run without committing the rest of the app to a particular modem.

A16. Assessment of the mentor's informal message

The message is useful as a requirements critique. Its core distinctions agree with the primary evidence reviewed here: external LoRa receivers, separate phone/radio coverage, foreground-first P2P, signed alerts, private reports, honest delivery states and a complete demonstration.

I would qualify or extend it in these ways:

- "Technically possible" applies to the components and a bounded system; it does not yet establish a reliable national service.
- Its range and approximately 1 GB storage figures are allocations to test. Require only a small essential package by default; make larger maps optional and pre-downloadable.
- Its 28-84 messages/hour calculation for 160 bytes at 10-30 bit/s is arithmetically reasonable after rounding. It assumes the stated rate already represents the relevant continuously available shared capacity. It is not a LoRa network-capacity measurement and cannot replace the airtime, duty-cycle and contention analysis.
- An ordinary Wi-Fi hotspot is a useful transport. Supplying a trusted app, maintaining its local state and solving permissions/HTTPS origin boundaries require additional implementation.
- A national alert cannot reach a completely disconnected local mesh without an independent feed or a physical carrier. Likewise, local publishing needs predelegated authority if cloud services are unavailable.
- A radio board's small price understates the costs of powered gateways, secure native software, maintenance and exercises. The minimum field kit is affordable; the verified service costs more.
- No signature scheme prevents jamming, establishes the truth of a citizen observation, or instantly revokes an unknown stolen key on an isolated phone.

The mentor's note is not treated as a primary source or evidence of measurements.

The next concrete decision is to fund the small end-to-end pilot and agree on an institutional message owner, one municipality, a permitted radio configuration, a broadcast-testing partner if available, and measurable delivery conditions. Scale the parts whose measured results justify it.

Appendix B. Latvian IDC response draft

Atbildes projekts Iekšlietu digitālajam centram

Sagatavots 2026. gada 8. septembrī. Pārskatāms melnraksts; nav nosūtīts.

Labdien, Artūr!

Paldies par jautājumiem. Prezentācijas sadaļa "Līmenis 3" apraksta iecerēto attīstības virzienu. Tās formulējumus nepieciešams precizēt, nošķirot pašreizējo tīmekļa platformu no vēl izstrādājamās un pārbaudāmās bezsakarū funkcionalitātes.

Mūsu ieteiktais risinājums ir lietotne ar iepriekš saglabātiem datiem, nelieli digitāli parakstīti atjauninājumi un vairāki to piegādes ceļi. Drošai darbībai būtiskākais ir skaidri noteikt, kurš drīkst publicēt oficiālu informāciju, kā tā tiek pārbaudīta telefonā un kā lietotājam tiek parādīts tās aktualitātes un piegādes statuss.

Pašreizējā koda pārskatā ir konstatēta sagatavotības materiālu bezsaistes saglabāšana, bet nav pilnas bezsaistes kartes vai šeit aprakstītās radio/P2P piegādes. Tam nepieciešams izstrādāt telefonā instalējamu lietotnes daļu ar vietējo datu glabātuvī un sakaru saskarnēm.

Radio tehnoloģija un uztveršana. Pilotam piedāvājam atsevišķas LoRa radioierīces Latvijas frekvenču izmantošanas nosacījumiem atbilstošā konfigurācijā. Parasts iPhone vai Android telefons pats LoRa signālu neuztver. Telefons sazinātos ar ārēju ierīci, izmantojot Bluetooth LE, vai ar koplietošanas datu mezglu, izmantojot vietējo Wi-Fi. Šim Wi-Fi savienojumam internets nav nepieciešams.

Prezentācijā norādītie aptuveni 10-30 bit/s nav jāinterpretē kā Atbalsts prototipā apstiprināts mērījums. LoRa nominālais pārraides ātrums un lietotnei faktiski piegādātais datu ātrums ir atšķirīgi lielumi. Rezultātu ietekmē radioiestatījumi, atļautais raidīšanas laiks, ziņojuma izmērs, atkārtojumi, traucējumi un retranslatori. Pilotā būtu jāmēra pilna, pārbaudīta ziņojuma piegādes laiks abos virzienos.

Dati radio skaņā. Kā papildu saņemšanas ceļu var pārbaudīt dzirdamu modema toņu pārraidīšanu radio programmas skaņā. Radio uztvērējs saņemtu apraidi, bet atvērta lietotne ar telefona mikrofona uztvertu skaņu, atkodētu datus un pārbaudītu parakstu. Šāda principa piemērs ir **TIVAR**^[13]. Tas vēl neapstiprina mūsu risinājuma uzticamību dažādos mūsdienu telefonos. Gandrīz ultraskaņas signālu pārraide parastā FM apraidē nebūtu pamata izvēle. Šis apraides ceļš ir vienvirziena; atbildei vajadzīgs cits savienojums.

Informācijas saturs un karte. Pa šauro radio kanālu paredzētu pārraidīt notikuma identifikatoru, versiju, laiku, teritoriju vai koordinātas, apdraudējuma līmeni, rīcības kodu, nelielu papildu tekstu un digitālo parakstu. Karte, instrukcijas un pamatobjekti tiktu saglabāti telefonā iepriekš. Ar radio varētu pievienot arī jaunu resursu punktu, mainīt tā statusu vai attēlot vienkāršu apdraudējuma zonu.

Attēlus un video pa šo kanālu nepiedāvājam pārsūtīt. Piemēram, pat 100 kB attēls pie 10-30 bit/s prasītu aptuveni 7-22 stundas ideālas nepārtrauktas pārraides. Pielikumus varētu glabāt telefonā un vēlāk nodot pa vietējo Wi-Fi vai augšupielādēt pēc savienojuma atjaunošanās. Precīzs bezsaistes kartes apjoms jānosaka pēc datu un detalizācijas izvēles; sākotnējais plānošanas diapazons detalizētai vektorkartei ir daži simti megabaitu.

P2P un darbība fonā. Lietotnes varētu apmainīties ar trūkstošajiem ziņojumiem, kad ierīces atrodas tuvumā. A nodotu ziņojumu B, savukārt B vēlāk to nodotu C. Pirmajai versijai paredzētu lietotāja aktivizētu apmaiņu ar atvērtu lietotni. Nevaram solīt nepārtrauktu automātisku retranslāciju visos iPhone: operētājsistēma ierobežo darbību fonā, un iespējas atšķiras atkarībā no izmantotās tehnoloģijas. **Apple skaidrojums par Bluetooth darbību fonā**^[26].

Pastāvīgai datu uzkrāšanai un nodošanai izmantotu īpašus mezglus ar nodrošinātu barošanu. Ziņojumu identifikatori, versijas, derīguma nosacījumi, saņemto datu uzskaitē un pārraides ierobežojumi novērstu normālu ierīču bezgalīgu atkārtošānu. Beidzies aktualitātes termiņš nenozīmētu "apdraudējums beidzies"; lietotne parādītu, ka jaunākais stāvoklis nav zināms.

Drošība un iedzīvotāju ziņojumi. Oficiālus paziņojumus parakstītu tikai pilnvarots izdevējs. Telefoni un parastie datu mezgli saņemtu pārbaudes atslēgas, nevis tiesības izveidot oficiālus paziņojumus. Tādēļ kompromitēts

retranslator pats nevarētu radīt derīgi parakstītu viltus trauksmi. Tas gan varētu aizturēt datus vai traucēt piegādi; pret to nepieciešami alternatīvi ceļi, pārraides ierobežojumi un uzraudzība.

Iedzīvotāja palīdzības pieprasījumu varētu šifrēt telefonā saņēmējam centrā. Starptiekierīces pārvadātu šifrētu saturu. Mobilais mezgls mugursomā vai transportlīdzeklī varētu apvienot radio uztvērēju, vietējo piekļuves punktu, glabātuvī un retranslatoru, kā arī vēlāk nogādāt savāktos datus serverī.

Lietotnē atsevišķi parādītu “saglabāts telefonā”, “nodots starptiekierīcei”, “saņemts centrā” un “izskatīts”. Nodošana tuvākajam mezglam pati par sevi neapstiprina pieprasījuma saņemšanu vai palīdzības nosūtīšanu.

Pārklājums un pilotprojekts. Viena mezgla rādīsim un vienlaikus apkalpojamo lietotāju skaitu var noteikt tikai konkrētai konfigurācijai un videi. Sākotnējai plānošanai starp LoRa ierīcēm var izmantot aptuveni 0,5-2 km sarežģītā pilsētvidē un 3-10 km laukos, bet šie nav izmērīti vai garantēti lielumi. Telefona savienojuma attālums līdz vietējam Wi-Fi/Bluetooth mezglam ir ievērojami mazāks.

Vispirms piedāvātu pārbaudīt vienu pašvaldību un konkrētus informācijas saņemšanas punktus. Nelielam demonstratoram ar sešām radioierīcēm, diviem datu mezgliem un aizlienētiem telefoniem aprīkojuma un lauka darbu plānošanas izmaksas būtu aptuveni 1 200-2 000 EUR bez PVN. Atsevišķi jāparedz lietotnes un protokola izstrāde: sākotnēji aptuveni 25-50 izstrādes cilvēkdienas. Tas vēl nebūtu publiskai operatīvai izmantošanai gatavs dienests.

Pilotā demonstrētu pilnu ķēdi: pilnvarots izdevējs izveido paziņojumu -> radio -> datu mezgls -> telefons -> notikums bezsaistes kartē -> otrs telefons -> iedzīvotāja pieprasījums -> serveris -> atpakaļ piegādāts saņemšanas apliecinājums. Paralēli pārbaudītu sakaru zudumu, atkārtotus un viltotus ziņojumus, darbību fonā, slodzi un rezerves barošanu.

Latvijas mērogā vispirms izvērtētu esošo apraidi un infrastruktūru, neatkarīgi pieslēgtus reģionālos mezglus un vietējos piekļuves punktus. Valsts mēroga pakalpojuma pieejamība būtu jānošķir no nepārtraukta radio pārklājuma katram telefonam. Pilnīgi izolētam vietējam tīklam jauni valsts līmeņa paziņojumi nevar piekļūt bez neatkarīga sakaru ceļa vai fiziskas datu nogādāšanas. Vietējo paziņojumu izveidei arī bez mākoņpakalpojumiem būtu vajadzīgs iepriekš pilnvarots pašvaldības izdevējs un atbilstoša atslēgu pārvaldība.

ePPO atsauce. To precizētu kā piemēru iedzīvotāju novērojumu nodošanai atbildīgajām institūcijām. Pārskatītie publiskie avoti neapstiprina prezentācijas apgalvojumu par video publicēšanu vai mūsu piedāvāto LoRa/P2P arhitektūru. Šos apgalvojumus no prezentācijas būtu jāizņem, kamēr nav atbilstoša pamatojuma. **ePPO oficiālā vietne** ^[40], **izstrādātāja lietotnes apraksts** ^[41].

Turpmākā sadarbībā būtu vērtīgi vienoties par pilotvietu, pilnvaroto informācijas izdevēju, pieejamo infrastruktūru un izmērāmiem piegādes kritērijiem. Tas ļautu pieņemt nākamās lēmumus, balstoties faktiskos rezultātos.

Source links

Technical sources accompany the original 8 September 2026 research. Funding and partner-service pages were checked during the subsequent planning discussion on the same date. Programme eligibility and deadlines should be rechecked before applying. Personal relationships are user-provided context, not independently verified affiliations or commitments.

Code labels identify the repository revision or local source file reviewed, not the revision currently deployed. Transfer times, radio airtime, idealised site counts, energy and budgets were checked using a local Python calculation script retained in the editable project materials. Those calculations use analytical assumptions and perform no radio or device test; the script is not a separate Drive upload.

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